

APARNA SHANKAR

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EDUCATION

The George Washington University

Master of Science in Data Analytics, **GPA: 3.63 / 4**

Relevant course work: Big Data, Statistics, Databases, Machine Learning, Data Visualization, Computer Vision, NLU, Algorithms

Aug 2023 - May 2025

Washington DC

Sri Ramachandra Faculty of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering – AI & ML, **GPA: 3.72 / 4**

Aug 2019 - June 2023

Chennai, India

SKILLS

Programming & ML frameworks: Python, R, Tensorflow, Scikit-learn, C, C++, Java, MATLAB, HTML, CSS, JavaScript

Data analysis & Databases: Tableau, Power BI, SPSS, Data Wrapper, PostgreSQL, MySQL, Splunk, Minitab

Computer software/ frameworks: React, Vue.js, Next.js, Node, GraphQL, AWS, Selenium, Figma, React Native, Firebase, Jira, Git

PROJECTS

Marvel Franchise Data Analysis | [Link](#)

Nov 2024 - Dec 2024

- Compiled data from sources, including box office revenue, critical reviews, and audience ratings, into unified dataset illustrating the decline of the Marvel franchise over the years 2008-2023.
- Designed charts with **DataWrapper** and integrated them into an **HTML** layout, highlighting an **85% drop** in opening weekend revenue and a **74% decline** in total box office revenue, reflecting audience disengagement and market saturation.

Driver Distraction Detection using Computer Vision| [Link](#)

Mar 2024 - Apr 2024

- Developed a **computer vision** using **CNN** model in **Keras** to classify driver activities (e.g., texting, talking on the phone) and detect distractions in real time, improving road safety achieving **98.93%** training accuracy and **98.26%** validation accuracy.
- Trained **YOLOv5** for in-cabin object detection and applied **data augmentation** to improve model robustness against lighting and occlusion variations.

Heart Stroke Prediction | [Link](#)

Jan 2022 - Mar 2022

- Automated clinical workflows to streamline heart stroke risk analysis and enhance decision-making for the healthcare sector.
- Trained a predictive model using **Machine Learning** algorithms like **Random Forest, SVM, and ensemble techniques**, achieving **94% accuracy** on benchmarked datasets for early risk detection.

RESEARCH PROJECTS

Assessment of Cardiac Dynamics and Risk Factor Analysis Using Deep Neural Nets

Jan 2022 - Mar 2022

- Co-authored a research paper on **cardiac dynamics** and stroke risk analysis using deep neural networks, **published in IGI Global**.
- Applied **machine learning** models along with feature selection and dimensionality reduction pipeline using LDA and SelectKBest methods to assess and predict heart stroke risks, contributing to the development of early detection that reduced stroke prediction errors by 15%.

WORK EXPERIENCE

Revizify Learning Solutions Pvt Ltd

Nov 2021 – Jan 2022

Frontend Developer Intern

New Delhi, India

- Built a platform using **Next.js** and **JavaScript** that enables users to create a personalized, flashcard-based revision system tailored to their needs.
- Implemented the frontend setup for the full application with a UI framework, achieving **20%** optimization in performance and successfully integrating **Django**-developed APIs with the user interface, resulting in **45%** user engagement.

Feminist Pen

Dec 2021 – Dec 2022

Software Developer Intern

Chennai, India

- **Collaborated** on the development of an internal **web application** for payroll management and employee records.
- Implemented a real-time **clock-in/out** system using **Firebase** and **Next.js**, enabling seamless employee attendance tracking and automated payroll integration, improving efficiency and reducing manual errors.

The Sparks Foundation

Mar 2021 – Apr 2021

Data Science Intern

Chennai, India

- Designed and implemented a supervised machine learning model to predict students' percentages and marks based on their study hours, aiming to provide data-driven insights for academic performance.
- Applied a **regression** model using the **Scikit-learn**, to predict student performance, achieving an RMSE of 6.23, effectively modeling the relationship between study hours and student performance.